Nicholas Pianetto

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EDUCATION

Iowa State University, College of Engineering **Expected December 2026**

Bachelor of Engineering, Computer Engineering

SKILLS

- **Programming Languages**: C, Java, Python, Verilog, VHDL.
- **Hardware Tools**: Oscilloscope, Spectrum Analyzer, Soldering Iron (THT, SMT assembly and rework), Desoldering Gun, Multimeter, Variable Power Supply, Logic Analyzer, Arduino, STM32, EPROM Programmer, Current Clamp.
- **Software**: IntelliJ, Visual Studio Code, QuestaSim, Quartus Prime, Android Studio, Git, GitHub, PADS Logic, PADS Layout, KiCad, STM32Cube IDE, Perforce, Lattice Diamond, FTDI Programmer (FTProg), SolidWorks, Linux, Zuken, Altium.

EXPERIENCE

Electrical Engineer Intern - *Vermeer* **May 2026 - Present**

- Designed and shipped a cable harness adapter to maintain pin compatibility for vendor part revision changes.
- Architected an automated wire harness tester, integrating CANbus, controllers, and a Linux-based display module for verification of machine harness pinouts.
- Managed vendor relations and component procurement, sourcing custom wire harness designs for different machines.
- Conducted field testing on a machine to monitor inrush current for cooling systems and mapped logged data against interconnect thermal limits to engineer a mitigation strategy to limit over-current failures.
- Designed a schematic in Altium for a Renesas RA4M1 Arm Cortex-M4-based system and produced a BOM sheet to provide an estimate on pricing.

Circuits II Teaching Assistant - *Iowa State University* **January 2026 - May 2026**

- Provided constructive feedback for students in the lab with hands-on troubleshooting, equipment use, and circuit synthesis.
- Aided in grading assignments and quizzes and provided helpful feedback for students.

Research Assistant - *Iowa State University* **May 2025 - May 2026**

- Independently designed both the schematics and board layout, validated functionality, and assembled an ESP32-based bioimpedance acquisition device utilizing the internal DFT from a high-precision impedance converter for animal research.
- Conducted meetings with the Animal Science department to establish design requirements such as built-in SPI SD card data logging and a portable LCD interface and wrote the firmware required to drive both features.

Embedded Design Engineer Intern - *Ag Leader Technology* **May 2024 - Dec 2024**

- Revised a serial to fiber optic PCB in PADS Layout and PADS Logic for use during ESD, EMC, and EMI testing.
- Collaborated with a team of embedded engineers to design the layout for a passive board in PADS while including proper EMI and ESD preventative design considerations.
- Reworked prototype PCBs by soldering experimental shielding and 0402-sized bypass capacitors in an effort to minimize capacitive coupling problems during ESD testing and successfully mitigated identified issues.
- Independently developed the schematic and board layout for a passive USB Type-C PCB with impedance matching of all differential pairs to achieve support of DisplayPort functionality for product testing. Initiated and led design review of the schematic and layout.

Computer Museum Volunteer - *Iowa State University* **March 2023 - Dec 2024**

- Collaborated with the Director of Cyber Security at Iowa State to help curate a computer museum highlighting Iowa State's extensive collection dating back to the first digital computer, the Atanasoff-Berry Computer, from 1937.

PROJECTS

PONG On Custom FPGA Development Board With Onboard VGA

- Independently designed schematics and board layout, fabricated, validated, and programmed a custom FPGA board based around Lattice Semiconductor's MachXO2 7000HC CPLD designed with on-board power regulation, configurable I/O voltage banks, synchronous VGA controller logic for 640x480 video timing, JTAG port, and expansion headers.
- Wrote a two-player PONG game in Verilog HDL with working controllers prototyped from salvaged component switches from existing machinery for quick proof of concept.
- Developed an FTDI FT2232H-based JTAG programmer for plug-and-play development.